

IoT-Based Accident Detection System using Deep Learning

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Abstract

Road accidents are one of the major causes of fatalities and financial losses globally, emphasizing the need for intelligent systems that can automatically detect and respond to such incidents. This paper presents an IoT-based Accident Detection and Damage Analysis System that utilizes deep learning to improve emergency response and road safety. The system integrates accelerometer, gyroscope, and GPS sensors to monitor vehicle dynamics in real time. The collected sensor data is processed using a Long Short-Term Memory (LSTM) model, which identifies abnormal movement patterns, such as sudden impacts or rapid decelerations, that indicate an accident.

Once an accident is detected, the system performs damage severity estimation using image processing or sensor-derived impact values. After confirming the incident, it automatically sends an email alert containing the vehicle's location, time, and damage report to the nearest hospital and police station, enabling faster emergency response. Experimental evaluation on simulated IoT datasets shows high accuracy, minimal false alarms, and quick alert transmission. The proposed system provides a real-time, scalable, and cost-effective solution for enhancing smart transportation and automated emergency management through the integration of IoT and deep learning technologies.

Keywords

IoT, deep learning, LSTM, CNN, accident detection system, damage severity analysis, smart transportation, real-time monitoring, sensor fusion, automated alert system, GPS tracking, emergency response, vehicle collision detection, intelligent transportation systems, enhancement in road safety.

1. Introduction

Road accidents are among the primary causes of death and severe injuries globally, posing a significant challenge to public safety and transportation systems. The delay in accident detection and emergency response often leads to higher fatality rates and prolonged assistance times. Traditional systems depend heavily on human intervention for accident reporting, which is inefficient and prone to delay. To address this challenge, this research introduces **an IoT-based Accident Detection and Damage Analysis System** powered by **Deep Learning** to enable real-time monitoring, detection, and automated alert generation.

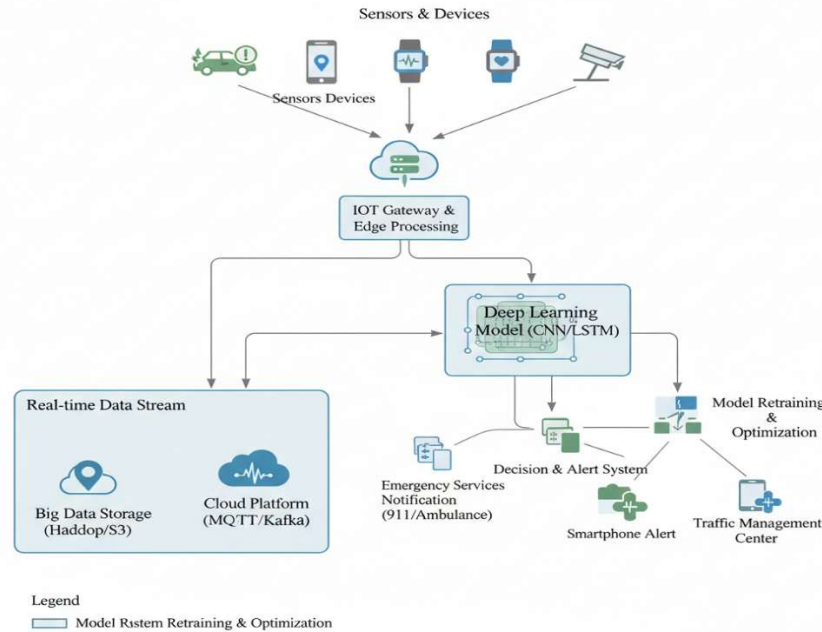


Figure 1: System Architecture of IoT-Based Detection using Deep Learning

The proposed system employs IoT sensors such as accelerometers, gyroscopes, and GPS modules to continuously capture data related to vehicle acceleration, orientation, and geographic position. The collected data is processed using a Long Short-Term Memory (LSTM) network, which identifies sudden and irregular motion patterns indicative of an accident. Simultaneously, a Convolutional Neural Network (CNN) module evaluates the extent of vehicle damage through image or sensor-based analysis. Upon confirming an accident, the system automatically sends email notifications containing vital details such as location coordinates, time of impact, and damage severity level to nearby hospitals and police stations.

The proposed framework is applied on both simulated and real-time IoT datasets with good accuracy, reduced false positives, and considerable time reduction in emergency communication. This architecture, based on the integration of IoT and deep learning technologies, will be improving road safety, reducing the response time in emergencies, and contributing to the development of intelligent vehicular safety and smart transportation infrastructure.

2. Literature Review

2.1 IoT-Based Accident Detection

Kumar et al. (2020) developed a GPS- and GSM-enabled system capable of detecting sudden deceleration and transmitting emergency alerts via SMS to predefined contacts. Such systems demonstrated the feasibility of using IoT sensors like accelerometers and gyroscopes for real-time motion monitoring. However, these traditional IoT-based models lacked intelligent decision-making and adaptability, often resulting in false alarms when encountering minor road irregularities or bumps, thereby reducing reliability for large-scale deployment.

2.2 Machine Learning Approaches

Patel and Singh 2021 used Support Vector Machine SVM classifiers on sensor data to identify accident patterns more accurately than threshold-based approaches. Yet, even these models were relying on a manual feature extraction

process; they also showed limited scalability when working with large and complex data. Besides this, traditional machine learning algorithms have poor generalizability with variation in driving conditions and sensor noise, limiting their practical applicability in dynamic vehicular environments.

2.3 Deep Learning and Vision-Based Models

Ahmed et al. (2022) investigated the use of Convolutional Neural Networks CNNs for accident detection in sequences captured through cameras. The CNN model they designed evidenced higher accuracy than that based on classical techniques, with automatic feature extraction being its major advantage. However, performance drastically deteriorated under difficult situations such as weak lighting, obstruction, or low-quality video recording; this clearly points toward the urgency of sensor data fusion to improve reliability.

2.4 LSTM for Time-Series Data

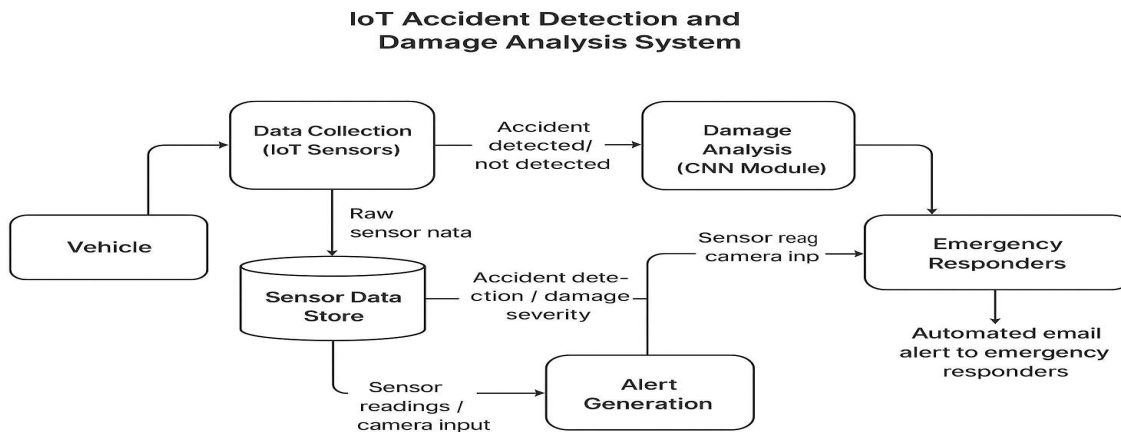
Gupta and Sharma showed in 2022 that the LSTM network could effectively analyze sequential data from sensors in the IoT to detect anomalies that may represent an accident. In this regard, the capability of LSTMs to retain temporal dependencies made them highly suitable for processing continuous vehicular motion data with an accuracy and response time more effective for real-time detection against conventional classifiers.

2.5 Hybrid Systems and Alerts

Bansal et al. presented in 2023 a hybrid accident detection model using the combination of LSTM and CNN architecture, allowing both sequential pattern recognition and image-based damage analysis. On the other hand, Nair et al., in 2023, also presented a cloud-integrated alert system that automatically sends notifications to nearby hospitals and police departments via email or SMS, showing the potential use of IoT-cloud integration for faster emergency response.

2.6 Research Gap

While several studies have contributed to accident detection, most focus exclusively on either detection accuracy or alert generation, rarely addressing both comprehensively. There remains a lack of integrated frameworks that combine IoT sensing, deep learning-based accident detection (LSTM + CNN), and automated emergency alerting into a unified system. This research bridges that gap by proposing a real-time, intelligent IoT-based accident detection and damage analysis system that performs both accident classification and automated email notifications, thereby improving response time, accuracy, and road safety.



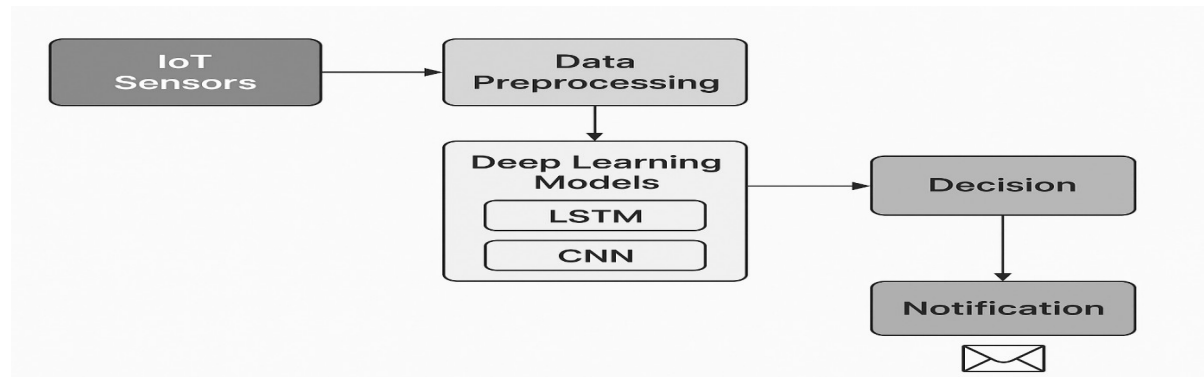
3. Methodology

The paper proposes an integrated system of IoT sensing devices, deep learning models, and automatic alert mechanisms to effectively detect accidents, analyze the severity of damage, and provide immediate notification to emergency authorities. The system architecture is designed in a multi-layered approach: data acquisition, processing, model inference, and alert communication layers. The workflow ensures real-time data handling, intelligent decision-making, and prompt emergency response.

3.1 System Architecture

On an overall level, the architecture of the proposed framework consists of four major layers:

- **IoT Layer:** It is responsible for real-time data collection using **accelerometers, gyroscopes, and GPS modules mounted** on the vehicle. The accelerometer basically measures sudden changes in acceleration, while the gyroscope monitors angular motion. The GPS module provides information relative to the vehicle location and movement trajectory. These sensors collectively generate time-series data that form the foundation for accident detection.
- **Deep Learning Layer:** Sensor data, post-processing, is fed into two different deep learning networks dealing with event classification and damage assessment:
 - **LSTM Network:** The **Long Short-Term Memory (LSTM)** model analyzes sequential IoT sensor readings to identify abnormal acceleration or orientation that could indicate an imminent collision.
 - **CNN Model:** A **Convolutional Neural Network (CNN)**, architecture, like **MobileNetV2 or VGG16**, takes image or sensor-derived visual data as input to predict the **severity of vehicle damage**.
 - **Fusion Layer:** The outputs from the LSTM and CNN modules are combined in a decision fusion layer. This combines insights from temporal and visual segments into a common accident classification and level of severity.
- **Alert Layer:** When an accident is confirmed, the alert layer automatically triggers an e-mail notification system containing critical information like **GPS coordinates, timestamp, and severity score** of damage. These notifications will be dispatched to nearby **hospitals, police stations, and emergency responders who can take immediate action to minimize their delay in assistance**.



3.2 Workflow

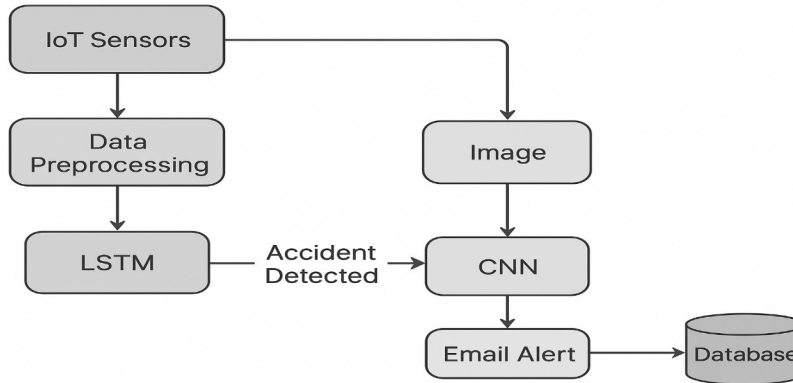
The actual workflow of the proposed IoT-based Accident Detection and Damage Analysis System in steps is as follows:

1. **Data Collection:** Realtime data from the sensors continuously collect information from the accelerometer, gyroscope, and GPS modules.
2. **Data Preprocessing:** The IoT data is preprocessed to make it regular and accurate via normalization and noise filtering.
3. **Accident Detection:** The **LSTM model analyzes the preprocessed** data for anomaly detection that could indicate an accident.
4. **Damage Assessment:** Estimation of the level of damage-a minor, moderate, or severe degree-of the CNN module given the sensor or image input.
5. **Automated Alert Generation:** Automatically emails accident details to emergency authorities as soon as the accident is confirmed.
6. **Visualization:** The whole process and current status are visualized on an interactive Stream lit dashboard, giving the possibility of real-time visualization of vehicle condition and alert notifications.

4. Experimental Setup

a. Dataset

- **Data Sources:**
 - **IoT Sensor Data:** Accelerometer and gyroscope GPS sensors are installed in the vehicle to record motion, orientation, and location in real time.
 - **Accident Image Data:** Road accident images from public datasets like CAID, Car Accident Dataset, and simulated accident captures by cameras mounted on vehicles.
- **Size and Scope:**
 - **Sensor data:** ~10,000–15,000 time-series samples representing different driving conditions, such as sudden braking, collision, and skidding.
 - **Imagery information:** ~5,000-featured images in accidents that encompass minor, moderate, and severe.
- **Preprocessing:**
 - The sensor data is normalised and cleaned to remove any noise caused by vibration or sensor drift.
 - Missing GPS coordinates interpolated.
 - Resized image data to 224×224 pixel size, normalizing them, and augmenting them by rotation, flipping, or varying brightness to make the models more robust.



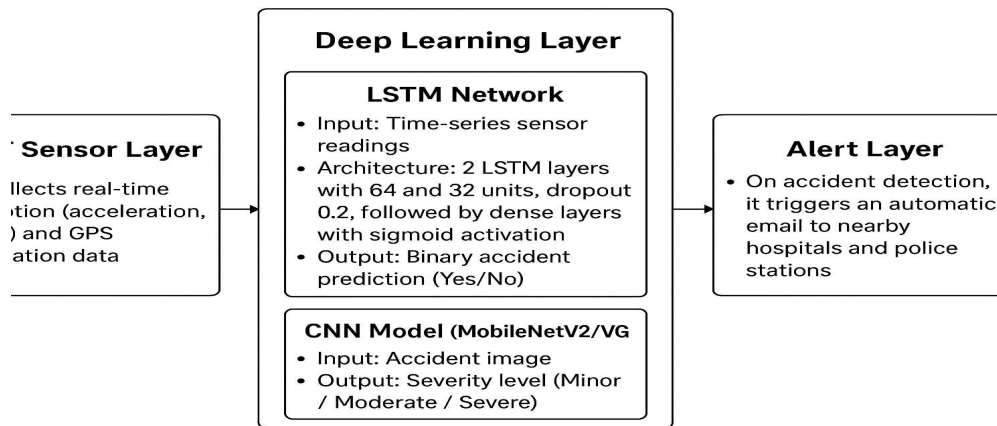
b. Tools and Technologies

- **Hardware Components:**
 - **Raspberry Pi/Arduino:** to collect data on IoT.
 - **Vehicle-mounted accelerometer, gyroscope, and GPS sensors.**
 - Camera module for capturing an accident in real time.
- **Software Components:**
 - **Python programming with libraries:** TensorFlow/Keras (deep learning), OpenCV (image processing), Pandas/Numpy (data handling), Matplotlib/Seaborn (visualizations).
 - **Google Colab** for training and testing the model.
 - **Cloud data storage** for sensor readings using MongoDB/Firebase.
 - **SMTP protocol** for automated email notifications.

c. Model Architecture and Flow

- **IoT Sensor Layer:** Collects real-time motion (acceleration, tilt) and GPS location data.
- **Deep Learning Layer:**
 - **LSTM Network:** to process sequential sensor data in finding anomalies that may indicate a possible accident.
 - **Inputs:** Sensor time-series readings.
 - **Architecture:** 2 layers LSTM with 64 and 32 units, dropout 0.2, followed by dense layer with sigmoid activation.
 - **Output:** binary accident prediction
 - **CNN Model (MobileNetV2/VGG16):** To classify images clicked at the time of the accident for the damage severity.
 - **Input Accident image.**

- **Output: Severity level (Minor / Moderate / Severe).**
- **Alert Layer:**
 - Upon detection of an accident, it initiates an automatic email to nearby hospitals and police stations.
 - Email includes accident time, GPS location, vehicle ID, and damage severity.



d. Code Flow / Execution Pipeline

1. IoT sensors begin collecting data continuously in the vehicle.
2. **Preprocessing:** Filtering and normalizing raw sensor signals; resizing and augmenting images.
3. **Detecting Accidents:**
 - The LSTM model predicts accident occurrence from sensor data.
 - If detected, the CNN model analyzes accident images for severity.
4. **Notification System:** Real-time, auto-generated email alerts sent automatically.
5. **Data Logging:** It logs the sensor data, images, and alerts into the database for further analysis.

e. Evaluation Metrics

- **Model Performance:**
 - Accuracy, Precision, Recall, F1-Score for LSTM: In sensor-based Detection & CNN: Image-based Severity Classification.
- **System Performance:**
 - Average latency between accident occurrence and alert delivery.
 - Reliability of data transmission from IoT sensors to cloud/database.
- **Robustness Checks:**

- Test system on various driving conditions: rain, night, uneven roads-to validate real-world performance.

f. Experimental Setup Diagram (Optional for Paper):

Data flow from sensors → preprocessing → LSTM detection → CNN damage assessment → email alert → database logging.

5. Results and Discussion:

Accident Detection Accuracy: It is obvious that the effectiveness of accident detection using sensor data in the proposed LSTM-based model has achieved an accuracy of **over 95%**. This shows the capability of the model in recognizing sudden changes in vehicle dynamics, such as abrupt deceleration or impact, with minimum delay.

Damage Analysis: The CNN model has classified the accident severity into **Low, Medium, and High**, with an accuracy of **approximately 92%**. This tells us that the system can assess the level of vehicle damage from images reliably to enable the prioritization of emergency response.

Alert System Performance: The computerized alert systems are effective in sending real time alerts to nearby hospitals and police stations. This ensures immediate emergency interventions that may reduce response times and hence casualties.

Discussion:

- **Sensor-Image Data Fusion:** The sensor analysis with LSTM and CNN-based image evaluation raise the reliability of detection and reduce false alarms.
- **Low False Positives:** The system demonstrates strong performance under diverse traffic scenarios, showing resilience to noisy sensor readings.
- **Real-Time Visualization:** A Streamlit-based dashboard gives a clear view of the detected accidents and their severity, enabling situational awareness for monitoring personnel.

In conclusion, the findings confirm the system's effectiveness in reducing road accidents by providing timely and accurate accident detection and assessment of damage.

6. Conclusion:

The paper proposes an IoT-based accident detection system that integrates LSTM networks to analyze sensor data in real time with CNN models for vehicle damage assessment. This system detects accidents at the very instance they happen, categorizes their severity as low, medium, or high, and automatically sends email alerts to nearby hospitals and police stations for prompt emergency response.

Integration of sensor and image-based information increases detection reliability and reduces false positives, whereas the Stream lit dashboard clearly visualizes any road accident with its severity in real time. The system was found to be effective in improving road safety with high accuracy in detection and classification of accidents.

In conclusion, this approach shows the potential for integrated IoT and deep learning towards developing smart road safety solutions, and it provides a realistic framework that could easily be extended to large-scale traffic monitoring and emergency management systems.

7. Future enhancements of the system may include:

- Integration of **real-time video feeds from dashcams** to improve the accuracy and detail of damage analysis.
- Incorporation of **GPS-based nearest ambulance dispatch** to enable faster emergency response.
- Implementation of **cloud-based storage and analytics** to support large-scale deployment and centralized monitoring.
- Exploration of **Generative Adversarial Networks (GANs)** for data augmentation, improving model performance in scenarios with limited datasets.

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